Casimir, CMB and H_0 — the shared ξ scale

Step-by-step derivation of the connection formula
with a consistency check of the SI conversion

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2026

Contents

What this is about

Three phenomena that at first have nothing to do with one another: the *Casimir effect* (vacuum force between plates, $\sim \mu$m), the *cosmic microwave background* (CMB, $T_{\text{CMB}} = 2{,}725$ K) and the *Hubble constant* H_0 (cosmic expansion rate). In FFGFT all three hang on *one* vacuum length scale $L_\xi \approx 100\,\mu$m and on the same exponent $41/4$ of the geometry parameter $\xi = 4/30000$. This note lays the connection open step by step (brought together in Doc. 166 from Doc. 091, 165, 025/061; status assessment per Doc. 190).

Throughout: $\xi = \dfrac{4}{30000} = 1{,}3333 \times 10^{-4}$.

.1 Preliminary remark: why "Hubble constant" is misleading

Before we use H_0, a terminological clarification – because the usual designation *Hubble constant* is, from the FFGFT view, misleading on two counts.

First: not a constant. Already in standard cosmology H_0 is not constant but the *present-day value* of the time-dependent Hubble parameter $H(t) = \dot{a}/a$. "Constant" there means only spatially homogeneous, not fixed in time. The name is thus already an established imprecision there.

Second (and sharper): in FFGFT there is no expansion rate at all. The static ξ universe has no scale factor $a(t)$ and no time-varying $H(t)$; the redshift is fractal path lengthening, not an expansion effect (Doc. 190, R14a/R15). There is therefore no "rate" whose present-day value one could take as H_0. The expansion reading is a ΛCDM pipeline interpretation (R18), not an FFGFT statement.

What remains physically is a length. The robust, model-neutral quantity is the *Hubble length*

$$R_H = \frac{c}{H_0}, \tag{1}$$

and $H_0 = c/R_H$ is merely this length expressed, via the SI conversion factor c, as an inverse time. R_H is a *reference length* – the upper scale of the cosmological torus sector – entirely analogous to the Planck length ℓ_P or to the system-specific bounds $R = \hbar/mc$. It is *not* the size of the universe and *not* the observable horizon (the latter $\approx 3\,R_H$ in ΛCDM); only a lower bound is measurable (Doc. 190, R36).

Terminology convention

In this note the cosmic scale is conceived primarily as a *reference length* R_H, not as a "constant" H_0. Where H_0 appears it is to be read as $H_0 = c/R_H$ – an SI restatement of the reference length, not a fundamental constant of nature. This is consistent with H_0 as an emergent, decompactified quantity (R16) and with the status of R_H as a shared empirical reference point (R39). Precisely for this reason the following derivation works throughout with *lengths* (L_ξ, R_H) and carries H_0 only as their conversion.

.2 Starting point: three relations from three documents

The three building blocks

1. **Casimir/vacuum (Doc. 091):** the CMB energy density as a T0 vacuum density,
$$\rho_{\text{CMB}} = \frac{\xi \hbar c}{L_\xi^4}. \tag{2}$$
2. H_0 **ratio (Doc. 165):**
$$\frac{\hbar H_0}{m_e c^2} = \frac{\pi}{2}\,\xi^{10}. \tag{3}$$
3. **CMB temperature (Doc. 025/061):** $T_{\text{CMB}} = 2{,}72548\,\text{K}$, brought in via the Stefan-Boltzmann law (Step 1).

Relations (2) and (3) were developed independently. The question: are they two aspects of the same ξ structure?

.3 Step 1: Casimir ↔ CMB via the scale L_ξ

1a – Equating the two energy densities

The standard Casimir energy density between two plates at separation d is

$$\rho_{\text{Casimir}}(d) = \frac{\pi^2\,\hbar c}{240\,d^4}. \tag{4}$$

At exactly the separation $d = L_\xi$ it relates to the vacuum density (2) as

$$\boxed{\frac{\rho_{\text{Casimir}}(L_\xi)}{\rho_{\text{CMB}}} = \frac{\pi^2 \hbar c/(240\,L_\xi^4)}{\xi \hbar c/L_\xi^4} = \frac{\pi^2}{240\,\xi} \approx 308} \tag{5}$$

Remarkable: inserting (2) into (5), ξ cancels and one recovers *exactly* the standard Casimir formula. The T0 notation is thus consistent with the textbook Casimir formula; L_ξ is the scale at which the vacuum and Casimir densities are of the same order.

1b – L_ξ from the CMB temperature

The CMB density is thermal (Stefan-Boltzmann):

$$\rho_{\text{CMB}} = \frac{\pi^2}{15}\,\frac{(k_B T_{\text{CMB}})^4}{(\hbar c)^3}. \tag{6}$$

Equating (2) and (6) and solving for L_ξ:

$$\frac{\xi \hbar c}{L_\xi^4} = \frac{\pi^2}{15}\,\frac{(k_B T_{\text{CMB}})^4}{(\hbar c)^3} \implies \boxed{L_\xi = \left[\frac{15\,\xi}{\pi^2}\right]^{1/4} \frac{\hbar c}{k_B T_{\text{CMB}}}} \tag{7}$$

Casimir exponent $1/4$

Numerically: $L_\xi = 100{,}24\,\mu$m (0.11 % from the scale determined directly from ρ_{CMB}). Decisive for what follows: from (7), ξ enters L_ξ *with the power* $1/4$,

$$L_\xi \propto \xi^{1/4}.$$

.4 Step 2: H_0 as a ξ power

Relation (3) solved for H_0/c (with the reduced Compton wavelength $\bar{\lambda}_e = \hbar/m_e c$):

$$\frac{H_0}{c} = \frac{\pi}{2}\,\xi^{10}\,\frac{m_e c^2}{\hbar c} = \frac{\pi}{2}\,\frac{\xi^{10}}{\bar{\lambda}_e}. \tag{8}$$

H_0 exponent 10

Inserted, this gives $H_0 \approx 66{,}8$ km/s/Mpc – within $0{,}9\,\%$ of the Planck value $67{,}4$ km/s/Mpc, without free parameters. Here ξ enters with the power 10:

$$H_0 \propto \xi^{10}.$$

.5 Step 3: Multiplication – the bridge formula

Now the product of the two scales L_ξ (7) and H_0/c (8). The powers of ξ add, $\hbar c$ cancels:

$$\begin{aligned}
L_\xi \cdot \frac{H_0}{c} &= \left[\frac{15\xi}{\pi^2}\right]^{1/4} \frac{\hbar c}{k_B T_{\text{CMB}}} \cdot \frac{\pi}{2}\,\frac{m_e c^2\,\xi^{10}}{\hbar c} \\
&= \frac{\pi}{2}\left[\frac{15}{\pi^2}\right]^{1/4} \frac{m_e c^2}{k_B T_{\text{CMB}}} \cdot \xi^{1/4}\,\xi^{10} \\
&= \frac{\pi}{2}\left[\frac{15}{\pi^2}\right]^{1/4} \frac{m_e c^2}{k_B T_{\text{CMB}}} \cdot \xi^{41/4}.
\end{aligned} \tag{9}$$

Connection formula Casimir $\leftrightarrow H_0$

$$L_\xi \cdot \frac{H_0}{c} = \underbrace{\frac{\pi}{2}\left(\frac{15}{\pi^2}\right)^{1/4}}_{K_{\text{geo}}=1{,}7441} \cdot \underbrace{\frac{m_e c^2}{k_B T_{\text{CMB}}}}_{=\,2{,}176\times 10^9} \cdot \xi^{41/4}. \tag{10}$$

Numerical check: $L_\xi \cdot H_0/c = 7{,}241 \times 10^{-31}$ – computed directly and via (10) algebraically identical (to $10^{-14}\,\%$). The derivation is *exact* – no approximation, no free parameters *between* the building blocks.

.6 The exponent $41/4$ and its decomposition

The exponent is the sum of the two individual exponents from Steps 1 and 2:

$$\boxed{\frac{41}{4} = \underbrace{10}_{\substack{H_0\ \text{exponent}\\ (3)}} + \underbrace{\frac{1}{4}}_{\substack{\text{Casimir exponent}\\ (7)}}} \tag{11}$$

The same exponent $41/4$ appears independently in Doc. 026 as $E_H = E_0\,\xi^{41/4}$ (Hubble energy scale). The decomposition $41/4 = 10 + 1/4$ is therefore not constructed by chance: the 10 carries the coupling electron mass → Hubble scale, the $1/4$ carries the Casimir scaling of the vacuum energy.

.7 Where does the SI conversion factor sit?

A legitimate counter-question: we are operating here at the *decompactified* level of the formulation – H_0 exists only after decompactification (Doc. 190, R16: $H_0 = E_H/\hbar$ contains the $\hbar$ that becomes real only then), and L_ξ is a *physical* length ($\approx 100\,\mu$m) built from $\hbar$, c, k_B. At this level an SI conversion factor *must* appear. It does – it merely sits hidden in two places, and in the exponent notation in a third.

Place 1: the factor cancels (legitimate)

Both scales are built with the same $\hbar c$:

$$L_\xi \propto \frac{\hbar c}{k_B T_{\mathrm{CMB}}}, \qquad \frac{H_0}{c} = \frac{\pi}{2}\,\xi^{10}\,\frac{m_e c^2}{\hbar c}. \tag{12}$$

In the product $\hbar c$ drops out – because $L_\xi \cdot H_0/c$ is at its core a *length ratio* ($L_\xi/\bar{\lambda}_e$ with the reduced Compton wavelength $\bar{\lambda}_e = \hbar/m_e c$), and the ratio of two lengths is unit-free. This is no trick: the same SI conversion $\hbar c$ builds *both* lengths, so it must cancel in the ratio.

Place 2: the factor survives (the actual anchor)

What remains after the cancellation is *not* purely geometric:

$$\boxed{\frac{m_e c^2}{k_B T_{\mathrm{CMB}}} = 2{,}176 \times 10^9} \tag{13}$$

This contains c^2 (mass → energy) and k_B (Kelvin → energy) – and it is a *ratio of two measured quantities*: electron rest energy to CMB temperature. *That* is the SI / empirical conversion factor. The bridge formula (10) is thereby geometric *form* ($K_{\mathrm{geo}}\,\xi^{41/4}$) **times** an SI anchor ($m_e c^2/k_B T_{\mathrm{CMB}}$) – *not* a pure geometric identity.

Place 3: in the exponent – $41/4$ vs. $63/4$

The factor shows up most cleanly in the exponent itself. The $41/4$ is referenced to the electron / CMB energy scale E_0, *not* to the Planck scale. Planck-referenced (Doc. 190, R35) the exponent is $63/4$, and

$$\boxed{\frac{63}{4} = \frac{41}{4} + \frac{11}{2}} \tag{14}$$

The $11/2$ *is* exactly the SI scale conversion factor $\ell_P \leftrightarrow E_0$ (Planck length vs. electron / CMB scale). Put differently: if one writes $\xi^{41/4}$ and calls it "purely geometric", one has tacitly absorbed ℓ_P into the exponent – exactly the guardrail R35 ("the SI conversion factor ℓ_P must not be absorbed into the exponent"). Which reference scale one makes explicit is a *choice*; the factor never disappears, it only changes location:

Reference scale	Exponent	SI factor explicit as
E_0 (electron/CMB)	$41/4$	ratio $m_e c^2/k_B T_{\text{CMB}}$
ℓ_P (Planck)	$63/4 = 41/4 + 11/2$	extra exponent $11/2$

Why we are necessarily decompactified here

This is not a matter of style: H_0 *does not exist* in the compact T^4 form – it contains the $\hbar$ that becomes real only upon decompactification (R16) – and L_ξ is a concrete μm length that cannot be formulated without $\hbar, c, k_B$. The SI factors are therefore not optional but *constitutive*: they are what makes the quantities involved physical quantities in the first place. At the compact level there is neither H_0 nor this bridge – the question of the SI factor does not even arise there.

Verdict of the consistency check

The bridge formula is *consistent* – but as a *decompactified, SI-anchored* relation, not as a compactified T^4 identity. The SI conversion factor appears exactly as expected: partly cancelled ($\hbar c$, legitimate as a length ratio), partly as the surviving empirical anchor ($m_e c^2/k_B T_{\text{CMB}}$), partly – in the exponent notation – as the $11/2$ between the E_0 and Planck references (14). This is fully in line with Doc. 190: R34 (SI conversions indispensable for the c-/time-power, $c = \hbar = 1$ destroys the structure), R16 (H_0 emergent/decompactified) and R35 (no absorption of ℓ_P into the exponent).

.8 Inversion: H_0 directly from the Casimir scale

Since L_ξ follows via (7) from ρ_{CMB} (i.e. from T_{CMB}), (10) can be solved for H_0:

$$\boxed{H_0 = \frac{\pi}{2}\left[\frac{15}{\pi^2}\right]^{1/4} \frac{c}{L_\xi}\,\xi^{41/4}} \tag{15}$$

This is the experimental lever (Doc. 166, §5): a *Casimir precision measurement* of the scale L_ξ fixes, via (15), a value for H_0 – an H_0 determination in the laboratory rather than in the sky.

.9 Complete number chain

Quantity	Expression	Value
ξ	$4/30000$	$1{,}3333 \times 10^{-4}$
T_{CMB}	Planck 2018	$2{,}72548$ K
L_ξ	$[15\xi/\pi^2]^{1/4}\,\hbar c/(k_B T_{\mathrm{CMB}})$	$100{,}24\,\mu$m
ρ_{CMB}	$\xi\hbar c/L_\xi^4$	$4{,}170 \times 10^{-14}$ J/m³
$\rho_{\mathrm{Casimir}}/\rho_{\mathrm{CMB}}$	$\pi^2/(240\,\xi)$	≈ 308
K_{geo}	$(\pi/2)(15/\pi^2)^{1/4}$	$1{,}7441$
$m_e c^2/(k_B T_{\mathrm{CMB}})$	–	$2{,}176 \times 10^9$
$\xi^{41/4}$	$41/4 = 10{,}25$	$1{,}91 \times 10^{-40}$
$\hbar H_0/(m_e c^2)$	$(\pi/2)\,\xi^{10}$	$1{,}896 \times 10^{-38}$
$L_\xi \cdot H_0/c$	$K_{\mathrm{geo}}\,(m_e c^2/k_B T_{\mathrm{CMB}})\,\xi^{41/4}$	$7{,}241 \times 10^{-31}$
H_0	(15)	$66{,}82$ km/s/Mpc

.10 Honest assessment of the status (Doc. 190)

The chain above is algebraically clean, but its *epistemic* status must remain clear (corrections register Doc. 190, entries R20, R32, R33, R35, R36, R39):

What is derived – and what is not

- **Form forward:** that the connection is a *dimensionless ξ ratio* (Casimir scale × Hubble scale = ξ power) is structurally forward. The *chaining* of the building blocks (9) is exact.
- **Exponent not forward (R20):** the value $41/4$ – already the partial exponent 10 from (3) – is *not* derived forward from the T⁴ geometry. The neat decomposition $41/4 = 10 + 1/4$ (11) is *structurally suggestive*, not a proof.
- **ξ^N guardrail (R35):** "$X = \xi^N$" counts as content only if N is derived forward; otherwise any magnitude can be written as a ξ power. The SI factor ℓ_P must not be tacitly absorbed into the exponent (cf. Place 3 above).
- **H_0 is emergent, not fundamental (R16/R32):** H_0 contains the $\hbar$ that becomes real only upon decompactification and does not exist in the compact T^4 form.
- **Shared reference point (R39):** the Planck length↔ R_H relation follows from *no* framework out of first principles; ΛCDM fixes it via the measured H_0. FFGFT adopts the established value for comparability – this is *not* an asymmetric disadvantage but a shared empirical reference point of both models.

Verdict of the assessment

The *chaining* is exact – a consistency relation among SI-anchored quantities (L_ξ, R_H, m_e, T_{CMB}), together with the experimental lever Casimir → H_0 (15). ℓ_P does *not* appear in the formula; via the E_0 reference it is *extrapolable* – as a consistency check, not a derivation: the leg $E_0 \leftrightarrow \ell_P$ $(11/2)$ is derived (the fine-structure exponent, see Outlook), the cosmic leg $(41/4)$ remains open (R20). It is therefore *not* a first-principles prediction of H_0.

.11 Outlook: the $11/2$, the Λ fine-tuning problem and the H_0 tension

The $11/2$ is a starting point, not a proof. The exponent $11/2$ in the split $63/4 = 41/4 + 11/2$ is not arbitrary: it coincides with the *forward-derived* fine-structure exponent,

$$\alpha = \xi \cdot E_0^2 = K \cdot \xi^{11/2}, \qquad E_0 = \sqrt{m_e m_\mu} \propto \xi^{9/4} \quad \text{(Doc. 011, 070, 202, 210)}, \tag{16}$$

and it is the *same* E_0 to which the cosmic exponent is referenced ($E_H = E_0\,\xi^{41/4}$, Doc. 026). Numerically E_0 sits exactly where the split requires: $E_0/E_{\text{Planck}} = \xi^{5{,}48}$ versus $11/2 = 5{,}5$ – a match to $\sim 0{,}5\,\%$ (precisely the rounding $15{,}72$ vs. $15{,}75$ from R35).

However: the α exponent is referenced to the MeV / lepton base (the $9/4$ scaling), the split $11/2$ to the Planck scale (the offset) – different reference bases, the same number. Whether this is *structurally identical* or a numerical hit in the dense ξ^N field is open; the guardrail (R35, Doc. 271) requires for "content" an isolated, predicted, precise value. The identity would be proven only by $E_0 = E_{\text{Planck}} \cdot \xi^{11/2}$ forward. *Recorded as a conjecture, not as evidence.*

The reverse question: why does ΛCDM's H_0 sit there? Were there a genuine, fixed connection $\ell_P \leftrightarrow R_H$, that would be a sharp statement about H_0. But ΛCDM does *not* choose H_0 from theory – it *measures* it (and contested: the H_0 tension ~67 vs. ~73); ℓ_P comes from the measured $G, \hbar, c$. The ratio is not harmless:

$$\frac{\ell_P}{R_H} \approx 1{,}2 \times 10^{-61}, \qquad \left(\frac{\ell_P}{R_H}\right)^2 \approx 10^{-122} \;\sim\; \frac{\rho_\Lambda}{\rho_{\text{Planck}}}. \tag{17}$$

The $(\ell_P/R_H)^2 \sim 10^{-122}$ *is* the cosmological constant problem – the largest unexplained fine-tuning in physics. ΛCDM "chooses" the value by fitting Λ to the observed acceleration, without a theoretical reason. FFGFT eliminates Λ (R17/R39) and does *not* inherit the fine-tuning – but relocates the same "why this scale" into the exponent $41/4$.

The test. If the $41/4$ became forward-derivable (together with the already-derived $11/2$ leg), FFGFT would *predict* ℓ_P/R_H – and hence H_0 – instead of importing it; it could even decide the H_0 tension (which side, 67 or 73). Precisely that would turn the conjecture into evidence (isolation/prediction/precision, Doc. 271). Until then it remains a suggestive restatement of a measured ratio, open.